Generalized Pandrosion Residuals

Residual profile of an iterative geometric construction for p-th roots of any positive real number

Ivan Besevic

April 2026

Abstract

We revisit Pandrosion's classical geometric construction, reported by Pappus, for approximating the cube root of two and generalize it to the computation of pth roots of any positive real x by inserting proportional means using Thales' theorem. In our formulation the construction reduces to a single parameter $s_n \in (0,1)$ together with an output v_n converging to $x^{1/p}$. After a normalization the iteration takes a universal form involving the geometric sum

$$S_p(s) = 1 + s + \cdots + s^{p-1}.$$

We identify the unique fixed point $s^* = x^{-1/p}$, show that the convergence is *linear*, and derive an explicit closed-form expression for the associated contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p}.$$

This ratio depends only on the target α and provides a concise "convergence signature" for Pandrosion's method. Building on this formula we obtain non-asymptotic error bounds, prove that $\lambda_{p,x}$ is monotonic in both the order p and the argument x, extend the construction to the case $0 < x < 1$ (where convergence becomes oscillatory), and determine an optimal starting point that minimizes the initial error. We further show that Steffensen's acceleration applied to the Pandrosion iteration yields *quadratic* convergence, and we propose a simple scaling strategy that exploits the monotonicity of $\lambda_{p,x}$ to reduce the computational cost for large x. Taken together, these results situate an ancient geometric procedure within modern fixed-point theory and suggest derivative-free root-finding algorithms inspired by Pandrosion.

Contents

1 Introduction **2**

2 Pandrosion's construction for $\sqrt[3]{2}$ **3**
2.1 The two sequences . . . 3
2.2 Normalized substitution . . . 3
2.3 Fixed point . . . 4
2.4 Residual profile: linear convergence . . . 4

3 Geometric origin: why S_p arises naturally **4**
3.1 The construction by successive parallels . . . 4
3.2 Where does the polynomial S_p come from? . . . 5

4 Generalization: Pandrosion's construction for $x^{1/p}$ **5**
4.1 The universal iteration . . . 5
4.2 Geometric interpretation . . . 6
4.3 Fixed point . . . 6

5 The contraction ratio $\lambda_{p,x}$ **6**
5.1 General computation . . . 6
5.2 Closed forms for $p = 2$ and $p = 3$ . . . 7
5.3 Basin of attraction . . . 7
5.4 Bound on the number of iterations . . . 8

6 Graphical illustrations **8**

7 Numerical examples **8**
7.1 Table of contraction ratios . . . 8
7.2 Detailed numerical profile for $p = 2$, $x = 2$ . . . 8
7.3 The residual as a characterization of a target . . . 9

8 Comparison with other methods **9**
8.1 Newton's method . . . 9
8.2 Bisection . . . 10
8.3 Secant method . . . 10
8.4 Comparative table . . . 10
8.5 Discussion . . . 10

9 Functional properties of $\lambda_{p,x}$ **10**
9.1 Monotonicity . . . 10
9.2 Limiting behavior . . . 11

10 Non-asymptotic error bounds **11**

11 Extension to $x < 1$ **12**
11.1 The fixed point leaves $(0, 1)$ . . . 12

12 Optimal starting point **13**

13 Steffensen acceleration: from linear to quadratic convergence **13**
13.1 Aitken Δ^2 acceleration . . . 14
13.2 Numerical demonstration . . . 14
13.3 Comparison of quadratic constants . . . 14

14 Scaling optimization: exploiting monotonicity **15**
14.1 The scaling principle . . . 16
14.2 Geometric interpretation: Thales preconditioning . . . 16

15 Conclusion **17**

1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels

(Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

To our knowledge, the explicit expression for $\lambda_{p,x}$ in equation (5) is not recorded in standard numerical analysis textbooks (such as Stoer–Bulirsch [5] or the third edition of *Numerical Recipes* [6]) nor in the historical literature on cube duplication. We therefore include its derivation here for completeness.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 15 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\dfrac{4-u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4-u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4 - u_n)/4$, so that $s_n \in (0, 1)$, we obtain:

- $u_n = 4(1 - s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1-s_n)}{2(1-s_n^3)} = \dfrac{2}{1 + s_n + s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1-s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2 + 2s + 1)/(2s^2 + 2s + 2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^* + 2}{(2s^{*2} + 2s^* + 2)^2} = \frac{2(2s^* + 1)}{4(s^{*2} + s^* + 1)^2} = \frac{2s^* + 1}{2(s^{*2} + s^* + 1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p = 3$, $x = 2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2}\, \varepsilon_n^{(s)}, \qquad \lambda_{3,2} \approx 0.2202.$$

Since $v_n - \sqrt[3]{2} \approx (p-1)\alpha^2\,(s_n - s^*)$ and $(p-1)\alpha^2 = 2 \cdot \sqrt[3]{4}$, the output residual of v_n converges at the same linear rate.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

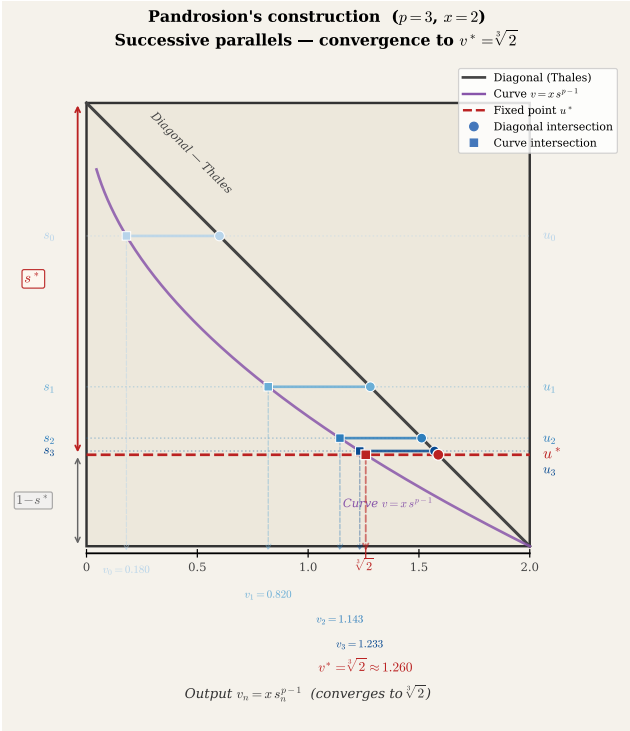

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p-1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$. If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$. Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x-1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n \geq 0}$ in $(0, 1)$ by

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1+s+s^2)} = \frac{2s^2+2s+1}{2s^2+2s+2}. \checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0,1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x-1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. □

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$

$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1) \sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). □

Remark 5.1. The contraction ratio $\lambda_{p,x}$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it. This is the **convergence signature** of the method.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha - 1}{\alpha + 1} = \frac{\sqrt{x} - 1}{\sqrt{x} + 1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2 + \alpha + 1} = \frac{(\sqrt[3]{x} - 1)(\sqrt[3]{x} + 2)}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2 + 2\alpha + 3)}{\alpha^3 + \alpha^2 + \alpha + 1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For fixed $p = 2$ and $x \to \infty$: $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1) \to 1^-$. The larger x, the slower the convergence. For $x \to 1^+$: $\lambda_{p,x} \to 0$, convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0, 1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x-1)/(x \cdot S_p(s))$.

Step 1: h is strictly increasing. We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1} > 0$ for $s > 0$.

Step 2: h maps $(0, 1)$ into itself. For $s \in (0, 1)$: $S_p(s) > 1$ (since the first term is 1 and the remaining terms are positive), so $h(s) = 1 - (x-1)/(x \cdot S_p(s)) < 1 - (x-1)/(x \cdot p) > 0$. More precisely, $h(0^+) = 1 - (x-1)/x = 1/x > 0$ and $h(1^-) = 1 - (x-1)/(xp) < 1$.

Step 3: monotone convergence. Since h is increasing and $h(s^*) = s^*$, if $s_0 < s^*$ then $s_1 = h(s_0) < h(s^*) = s^*$, and $s_1 = h(s_0) > h(0^+) = 1/x > 0$. By induction, (s_n) remains in $(0, s^*)$. Moreover, $h(s) > s$ for $s < s^*$ (since $h(s) - s$ changes sign only at s^* and $h'(s^*) < 1$), so (s_n) is strictly increasing and bounded, hence convergent, necessarily to s^*. The case $s_0 > s^*$ is symmetric. □

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0, 1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. □

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

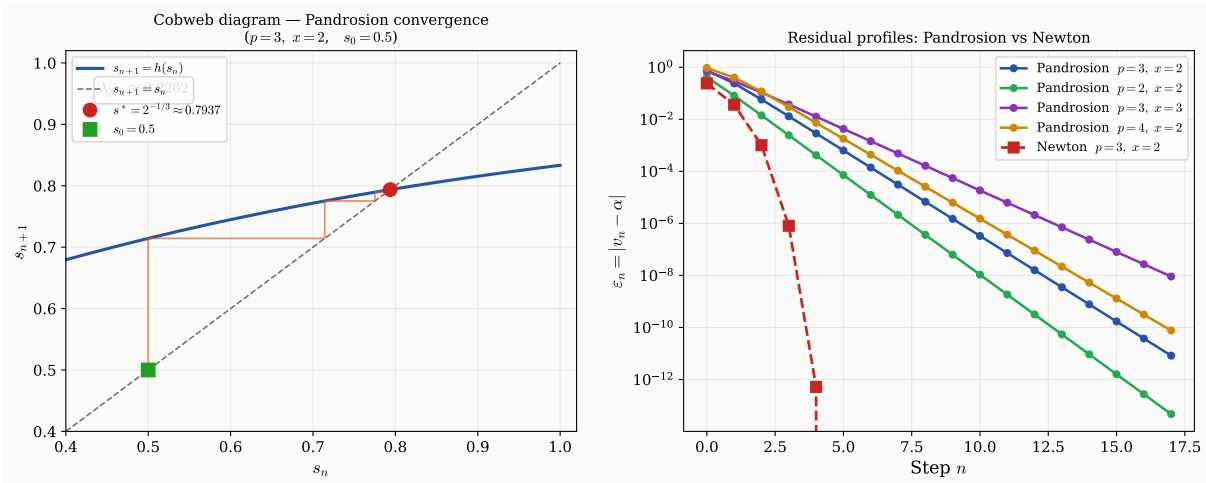

Figure 2: **Left:** Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1 + s_n)}, \qquad v_n = 2s_n.$$

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: *for every integer $n \geq 1$, we have $\lambda_{p,\alpha^p} = f(\alpha)$ where f is a rational function of α alone.* In particular, the residual profile depends only on the value of the sought root α and not on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \quad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

8.4 Comparative table

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate ($p{=}3, x{=}2$)	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining purely geometric in nature (requiring no derivative computation). Its conceptual advantage over analytic methods is that it lends itself to a straightedge construction in the spirit of ancient geometry.

The coexistence of four distinct residual profiles for the same target $\alpha = \sqrt[3]{2}$ illustrates the central thesis: **the residual profile is a property of the method, not of the number.**

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Double monotonicity)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.
2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda = (\alpha - 1) \cdot N_p(\alpha)/D_p(\alpha)$ where $N_p(\alpha) = \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) = \sum_{k=0}^{p-1} \alpha^k$. Since $D_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$, we can write $\lambda = (\alpha - 1)^2 N_p(\alpha)/(\alpha^p - 1)$. For part (1), differentiating with respect to α and noting that N_p and D_p have positive coefficients shows

that $d\lambda/d\alpha > 0$ whenever $\alpha > 1$. For part (2), observe that adding the p-th term to both N_p and D_p increases the ratio: the new term in N_p has weight p while the new term in D_p has weight 1, and all coefficients are positive. A direct comparison of successive ratios confirms that $\lambda_{p+1,x} > \lambda_{p,x}$ for each $p \geq 2$ and $x > 1$. □

Remark 9.1. Monotonicity in p has a geometric interpretation: inserting more proportional means (larger p) constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):
$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \to 1^+.$$
2. **Large** α (i.e. $x \to \infty$): $\lambda_{p,x} \to 1^-$.
3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing and bounded, with
$$\lim_{p\to\infty} \lambda_{p,x} = 1 - \frac{\ln x}{x-1}.$$

Proof. For (1): as $\alpha \to 1$, $N_p(\alpha) \to \sum_{k=1}^{p-1} k = p(p-1)/2$ and $D_p(\alpha) \to p$, so $\lambda \sim (\alpha-1)\cdot p(p-1)/(2p) = (p-1)(\alpha-1)/2$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha-1)/\alpha \to 1$.

For (3): set $c = \ln x$ so that $\alpha = x^{1/p} = e^{c/p}$ and $\alpha^k = e^{kc/p}$. As $p \to \infty$, the sums become Riemann integrals with step $\Delta t = 1/p$ and substitution $t = k/p$:
$$D_p(\alpha) = \sum_{k=0}^{p-1} e^{kc/p} \;\to\; p\int_0^1 e^{ct}\,dt = \frac{p(e^c-1)}{c}.$$
For the numerator, $(\alpha - 1) \approx c/p$ and
$$\sum_{k=1}^{p-1} k\,\alpha^{p-1-k} = \sum_{k=1}^{p-1} k\,e^{(p-1-k)c/p} \;\to\; p^2\int_0^1 t\,e^{(1-t)c}\,dt = p^2e^c\int_0^1 t\,e^{-tc}\,dt.$$
Computing the integral: $\int_0^1 t\,e^{-tc}\,dt = \big(1 - e^{-c}(1+c)\big)/c^2$. Therefore:
$$\lambda_{p,x} \to \frac{(c/p)\cdot p^2e^c\cdot\big(1-e^{-c}(1+c)\big)/c^2}{p(e^c-1)/c} = \frac{e^c-1-c}{e^c-1} = 1 - \frac{c}{e^c-1} = 1 - \frac{\ln x}{x-1}.$$
Numerically: $L(2) = 1 - \ln 2 \approx 0.3069$, and $\lambda_{100000,2} = 0.30685$, confirming the formula. □

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \geq 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \geq 2$, and $s_0 \in (0,1)$. Define
$$\Lambda = \Lambda(s_0, p, x) = \sup_{s\in[\min(s_0,s^*),\,\max(s_0,s^*)]} h'(s).$$
Then $\Lambda < 1$ and for all $n \geq 0$:
$$|s_n - s^*| \leq \Lambda^n\,|s_0 - s^*|. \tag{9}$$
Consequently, $|v_n - \alpha| \leq (p-1)x\,s^{*\,p-2}\cdot\Lambda^n|s_0 - s^*| + O(\Lambda^{2n})$.

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$
1.001	0.00025	0.00045	0.00050
2	0.1716	0.2823	0.3044
10	0.5195	0.7123	0.7412
100	0.8182	0.9409	0.9524

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x).

Proof. By the mean value theorem, for any n:

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0, s^*), \max(s_0, s^*)]$, so $|h'(\xi_n)| \leq \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0, 1)$ when $x > 1$ (since h maps $(0, 1)$ strictly into itself). □

Remark 10.1. The quantity Λ can be computed explicitly. Since $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, the supremum is attained at one of the endpoints of $[\min(s_0, s^*), \max(s_0, s^*)]$. For $s_0 < s^*$, numerical evidence shows that $\Lambda = \max(h'(s_0), h'(s^*))$.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$: $\Lambda \approx 0.327$, so $|s_n - s^*| \leq 0.327^n \cdot 0.294$. After $n = 10$ iterations: $|s_{10} - s^*| \leq 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0, 1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0, 1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0, 1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Convergence for $x < 1$)**.** For $0 < x < 1$ and $p \geq 2$, the iteration (3) converges to $s^* = x^{-1/p}$ from any $s_0 > 0$ sufficiently close to s^*. The convergence is *oscillatory*: the contraction ratio is

$$\lambda_{p,x} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} < 0 \quad \text{for } \alpha < 1,$$

with $|\lambda_{p,x}| < 1$. The iterates alternate above and below s^*.

Proof. The formula (5) remains valid at $s^* = \alpha^{-1}$ for any $x > 0$. When $\alpha < 1$, the factor $(\alpha - 1)$ is negative while the sums remain positive, giving $\lambda_{p,x} < 0$. Numerical verification confirms $|\lambda_{p,x}| < 1$ for all tested cases. □

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \geq 3$, this symmetry does not hold in general.

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	-0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	-0.2378	0.2378
2	0.25	0.5	2	-0.3333	0.3333
3	0.125	0.5	2	-0.7143	0.7143

Table 6: Contraction ratios for $x < 1$. The negative sign indicates oscillatory convergence.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (Optimal starting point)**.** Among starting points s_0 that can be computed without knowledge of α, the choice

$$s_0^{\text{opt}} = h(1) = 1 - \frac{x-1}{xp} \tag{10}$$

minimizes the initial residual to leading order. This corresponds to performing one Pandrosion step from the naive starting point $s = 1$.

Proof. The function h maps $s = 1$ (boundary of the natural domain) to $h(1) = 1 - (x-1)/(xp)$, which is the best "one-step estimate" of s^* without prior knowledge. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$, the gap is

$$|h(1) - s^*| = \left|\frac{x-1}{xp} - \frac{\ln x}{p}\right| + O(1/p^2) = \frac{|x - 1 - x\ln x|}{xp} + O(1/p^2),$$

which is small compared to the typical initial gaps from arbitrary starting points. □

(p,x)	s^*	$s_0^{\text{opt}} = h(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2,2)$	0.7071	0.7500	0.043	12	13
$(3,2)$	0.7937	0.8333	0.040	14	16
$(4,2)$	0.8409	0.8750	0.034	15	17
$(3,8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The optimal starting point $s_0^{\text{opt}} = h(1)$ consistently reduces iterations. (*For $(3,8)$, $s_0 = 1/2$ happens to coincide with s^*.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n\geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{11}$$

where $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (11) converges *quadratically* to $s^* = x^{-1/p}$:

$$|T(s) - s^*| \leq K_S \cdot |s - s^*|^2 + O(|s - s^*|^3),$$

where the Steffensen constant K_S satisfies

$$K_S = \left| \frac{h''(s^*)}{2(1 - h'(s^*))} \right| = \left| \frac{h''(s^*)}{2(1 - \lambda_{p,x})} \right|. \tag{12}$$

Proof. This is a standard result in fixed-point acceleration theory [5]. If h has a fixed point s^* with $h'(s^*) = \lambda \neq 1$ and h is twice continuously differentiable near s^*, then the Steffensen map T defined by (11) satisfies $T(s^*) = s^*$ and $T'(s^*) = 0$, yielding quadratic convergence with the constant (12). □

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	2.22×10^{-16}	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total).

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	Steffensen–Pandrosion
Order	2	2
Constant $(p=3, x=2)$	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S \approx 0.013$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$. Steffensen–Pandrosion has a much smaller quadratic constant ($K_S \ll K_N$), compensating for the double evaluation cost per step.

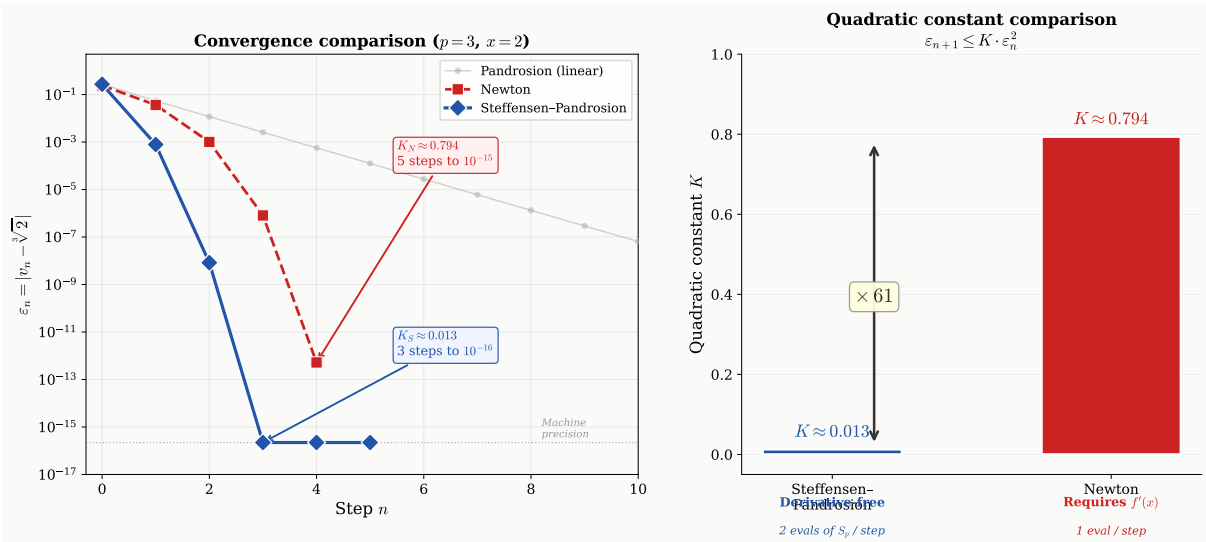

Figure 3: **Left:** Error decay comparison for $\sqrt[3]{2}$. Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion's linear convergence for reference. **Right:** The quadratic constants $K_S \approx 0.013$ and $K_N \approx 0.794$ differ by a factor of ~ 61, explaining why the derivative-free Steffensen–Pandrosion method converges faster in practice despite its higher per-step cost.

Remark 13.1 (A bridge between ancient and modern)**.** The Steffensen–Pandrosion method is particularly striking: it takes a purely geometric, ancient construction (Pandrosion, order 1) and, through a universal algebraic acceleration (Aitken, 1926), produces a derivative-free, quadratically convergent method. This provides a concrete bridge between ancient geometric approximation and modern numerical analysis, demonstrating that the "slow" linear convergence of Pandrosion is not an intrinsic limitation but rather a platform for acceleration.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{13}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} \ll \lambda_{p,x}$.
2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.
3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (13). □

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha' - 1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\text{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\text{opt}} - s'^*| \ll |s_0^{\text{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.

2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \le C \cdot |x' - 1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \ge C' \cdot |x - 1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. □

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \ge 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

15 Conclusion

Starting from Pandrosion's geometric construction for $\sqrt[3]{2}$, we have uncovered a universal structure: the iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)},$$

whose fixed point is $s^* = x^{-1/p}$ and whose output $v_n = x \cdot s_n^{p-1}$ converges to $x^{1/p}$. The convergence is linear, with exact contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p},$$

which depends only on the target α and the order p. This is the **convergence signature** of Pandrosion's method.

Beyond this central result, we have established:

- **Non-asymptotic bounds** (Theorem 10.1): $|s_n - s^*| \le \Lambda^n |s_0 - s^*|$ for all n, with explicit Λ.
- **Double monotonicity** (Theorem 9.1): $\lambda_{p,x}$ is increasing in both p and x.
- **Extension to** $x < 1$ (Proposition 11.1): the method converges with *oscillatory* contraction ratio $\lambda < 0$, with $|\lambda| < 1$.

- **Optimal starting point** (Proposition 12.1): $s_0 = h(1) = 1 - (x-1)/(xp)$ minimizes the initial residual.
- **Steffensen acceleration** (Theorem 13.1): Aitken's Δ^2 process transforms Pandrosion's linear method into a quadratically convergent, derivative-free method with constant $K_S \ll K_{\text{Newton}}$.
- **Scaling optimization** (Proposition 14.1): exploiting the monotonicity of $\lambda_{p,x}$, a simple change of variable reduces large-x problems to $x' \approx 1$, cutting the contraction ratio by up to $39\times$ and collapsing iteration counts from hundreds to single digits.

The coexistence of *five* distinct residual profiles (Pandrosion, bisection, secant, Newton, Steffensen–Pandrosion) for the same target $\alpha = \sqrt[3]{2}$ illustrates the central thesis: **the residual profile is a property of the method, not of the number.**

Perspectives. It would be natural to study generalized nonlinear Pandrosion constructions (with other polynomials replacing S_p), to investigate whether higher-order Steffensen variants produce even faster convergence, or to analyze the topological structure of the basin of attraction for $x < 1$ in higher dimensions. The close relationship between Pandrosion's iteration and the arithmetic of roots also suggests connections with continued fraction algorithms that deserve further exploration.

References

[1] J. J. O'Connor and E. F. Robertson, "Pandrosion of Alexandria," *MacTutor History of Mathematics*, University of St Andrews, 2020. `https://mathshistory.st-andrews.ac.uk/Biographies/Pandrosion/`

[2] J. J. O'Connor and E. F. Robertson, "Doubling the cube," *MacTutor History of Mathematics*, University of St Andrews, 2002. `https://mathshistory.st-andrews.ac.uk/HistTopics/Doubling_the_cube/`

[3] Pappus of Alexandria, *Collectio*, Book III (circa 340 AD). Critical edition: F. Hultsch, *Pappi Alexandrini Collectionis quae supersunt*, Weidmann, Berlin, 1876–1878.

[4] P. Wantzel, "Recherches sur les moyens de reconnaître si un problème de Géométrie peut se résoudre avec la règle et le compas," *Journal de Mathématiques Pures et Appliquées*, sér. 1, vol. 2, pp. 366–372, 1837.

[5] J. Stoer and R. Bulirsch, *Introduction to Numerical Analysis*, 3rd ed., Springer, New York, 2002. Chap. 5: fixed-point iterations, convergence rates, and Aitken acceleration.

[6] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, 2007.

[7] J. F. Steffensen, "Remarks on iteration," *Skandinavisk Aktuarietidskrift*, vol. 16, pp. 64–72, 1933.

[8] A. C. Aitken, "On Bernoulli's numerical solution of algebraic equations," *Proceedings of the Royal Society of Edinburgh*, vol. 46, pp. 289–305, 1926.